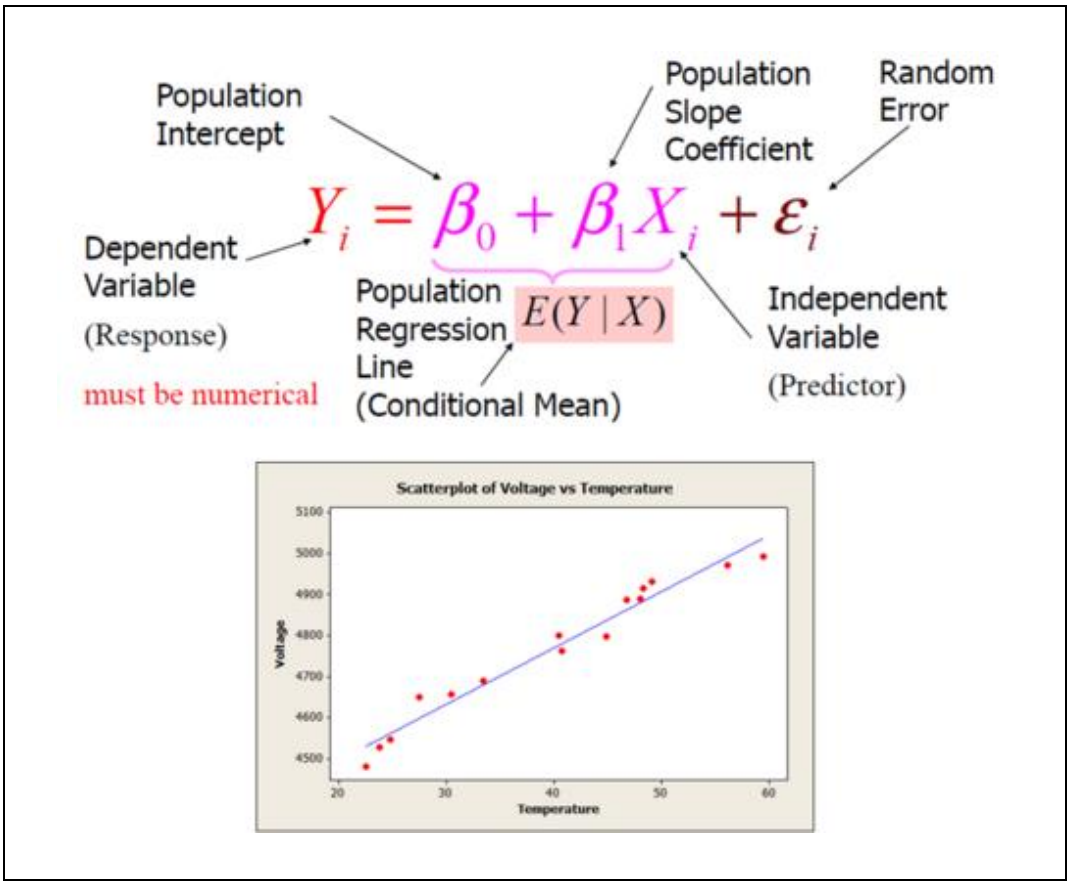
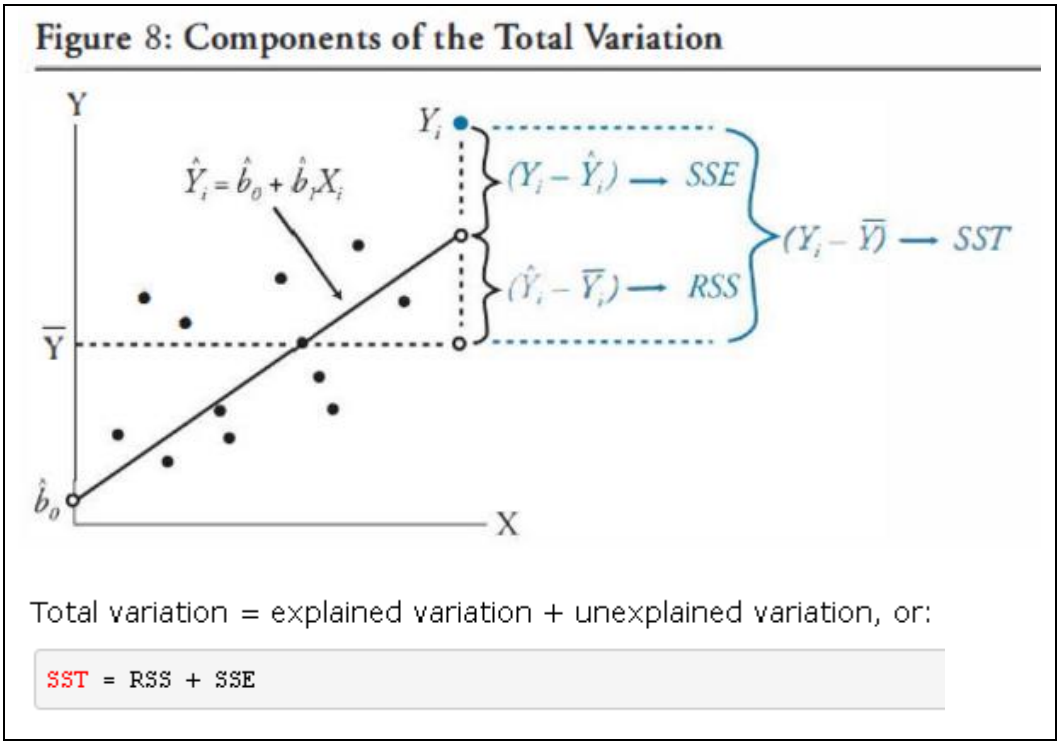
F79MB Chapter 5a Regression





Total variation = explained variation + unexplained variation, or:

```
SST = RSS + SSE
```

Coefficient of determination

Coefficient of determination, in statistics, R^2 (or r^2), a measure that assesses the ability of a model to predict or explain an outcome in the linear regression setting.

More specifically, R^2 indicates the **proportion of the variance in** the dependent variable (***Y***) that is predicted or **explained by linear regression and the predictor variable** (***X***, also known as the independent variable).
An R^2 of 0.35, for example, indicates that 35% of the variation in the outcome has been explained just by predicting the outcome using the covariates included in the model.

Example: We want to understand how well study hours predict exam scores. Let's assume our linear regression model fits the data well and we obtain an R^2 value of 0.80.

R^2 = 0.80 means that 80% of the variability in exam scores can be explained by the number of study hours. In other words, 80% of the variation in exam scores is accounted for by the variation in study hours.

The remaining 20% of the variability in exam scores is due to other factors not included in our model, such as innate ability, test anxiety, etc.

A higher R^2 value (closer to 1) indicates that the model explains a larger proportion of the variability in the dependent variable, while a lower R^2 value (closer to 0) indicates that the model explains less of the variability.

Section Chapter 5b: Model Diagnostics- Residuals and Influence

Anscombe's examples

- Frank Anscombe devised 4 sets of X-Y pairs
- He performed simple linear regression for each data set
- Here are the results

```
> data(anscombe)
> anscombe
   x1 x2 x3 x4    y1   y2    y3    y4
1  10 10 10  8  8.04 9.14  7.46  6.58
2   8  8  8  8  6.95 8.14  6.77  5.76
3  13 13 13  8  7.58 8.74 12.74  7.71
4   9  9  9  8  8.81 8.77  7.11  8.84
5  11 11 11  8  8.33 9.26  7.81  8.47
6  14 14 14  8  9.96 8.10  8.84  7.04
7   6  6  6  8  7.24 6.13  6.08  5.25
8   4  4  4 19  4.26 3.10  5.39 12.50
9  12 12 12  8 10.84 9.13  8.15  5.56
10  7  7  7  8  4.82 7.26  6.42  7.91
11  5  5  5  8  5.68 4.74  5.73  6.89
```

Means & Standard Deviations

Data Set	N	X		Y	
		Mean	SD	Mean	SD
1	11	7.50	2.03	9.00	3.32
2	11	7.50	2.03	9.00	3.32
3	11	7.50	2.03	9.00	3.32
4	11	7.50	2.03	9.00	3.32

The means and SDs for the 4 data sets are identical to two decimals.

Correlations between X and Y

Data Set	Pearson r	R-squared	Adj. R-sq	SE
1	0.82	0.67	0.63	1.24
2	0.82	0.67	0.63	1.24
3	0.82	0.67	0.63	1.24
4	0.82	0.67	0.63	1.24

Correlations, *R*-squared, adjusted *R*-squared, and standard errors are all identical to two decimals.

ANOVA Summary Tables

Data Set	Source	SS	df	MS	F	p
4	Regression	27.490	1	27.490	18.003	0.002
	Residual	13.742	9	1.527		
	Total	41.232	10			
3	Regression	27.470	1	27.470	17.972	0.002
	Residual	13.756	9	1.528		
	Total	41.226	10			
2	Regression	27.500	1	27.500	17.966	0.002
	Residual	13.776	9	1.531		
	Total	41.276	10			
1	Regression	27.510	1	27.510	17.990	0.002
	Residual	13.763	9	1.529		
	Total	41.273	10			

The Regression Coefficients

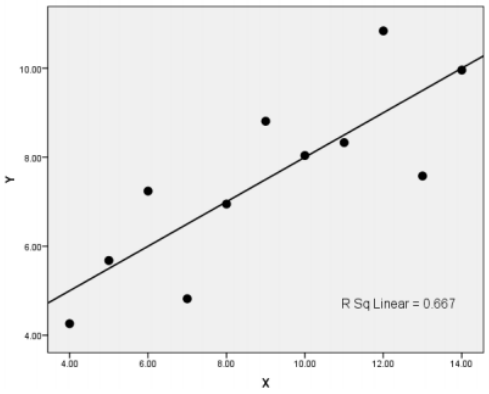
Data Set		B	SE	t	p	95% CI	
						Lower	Upper
4	Constant	3.00	1.124	2.67	0.026	0.459	5.544
	X	0.50	0.118	4.24	0.002	0.233	0.766
3	Constant	3.00	1.124	2.67	0.026	0.459	5.546
	X	0.50	0.118	4.24	0.002	0.233	0.766
2	Constant	3.00	1.125	2.67	0.026	0.455	5.547
	X	0.50	0.118	4.24	0.002	0.233	0.767
1	Constant	3.00	1.125	2.67	0.026	0.456	5.544
	X	0.50	0.118	4.24	0.002	0.233	0.767

For all 4 models, $Y' = 0.5(X) + 3$

Which Model is Best?

- Judging by everything we've just seen, it appears that the models are all equally good
- It is well known that good graphs are an essential part of data analysis (Tukey, 1977; Tufte, 1997)
- Let's look at some graphs that show the relationship between X and Y

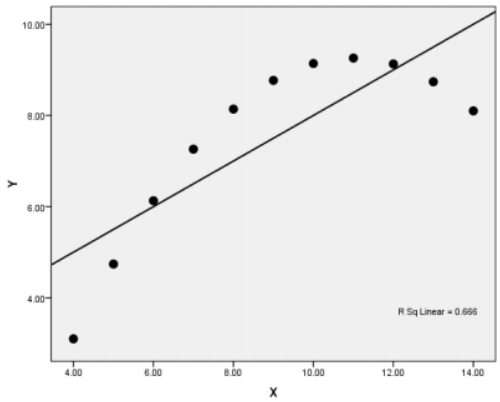
Scatter-plot for Data Set 1



This is a good looking plot.

No influential points; straight line provides a good fit.

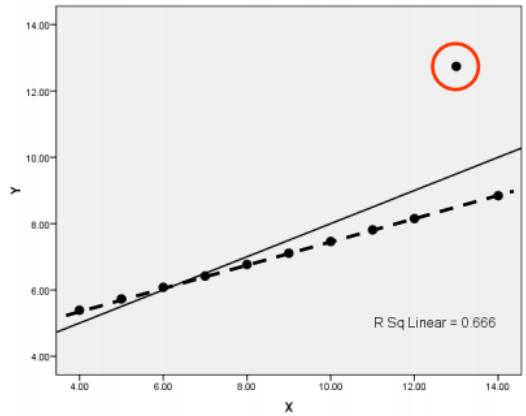
Scatter-plot for Data Set 2



Wrong model! The relationship between X and Y is ***curvilinear***, not linear!

The model should include both X and X^2 as predictors.

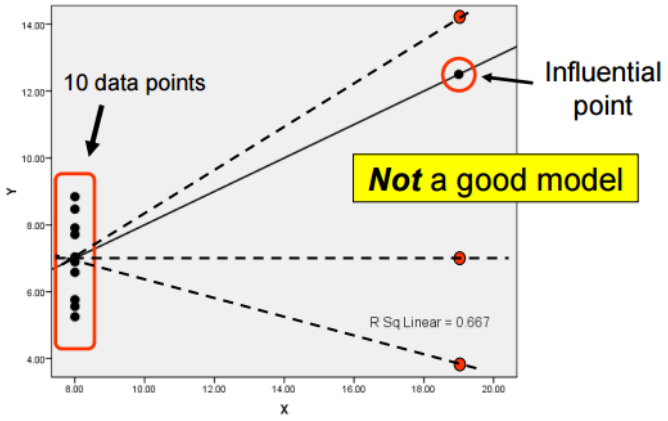
Scatter-plot for Data Set 3



Perfect linear relationship except for one **outlier**

Better model than for Data Set 1, but still not great.

Scatter-plot for Data Set 4



Summary

- The usual summary statistics for the 4 regression models were virtually identical
- Scatter-plots revealed that only one of the 4 data sets gave us a good model
- Appropriate graphs are an essential part of data analysis

Residual Plots

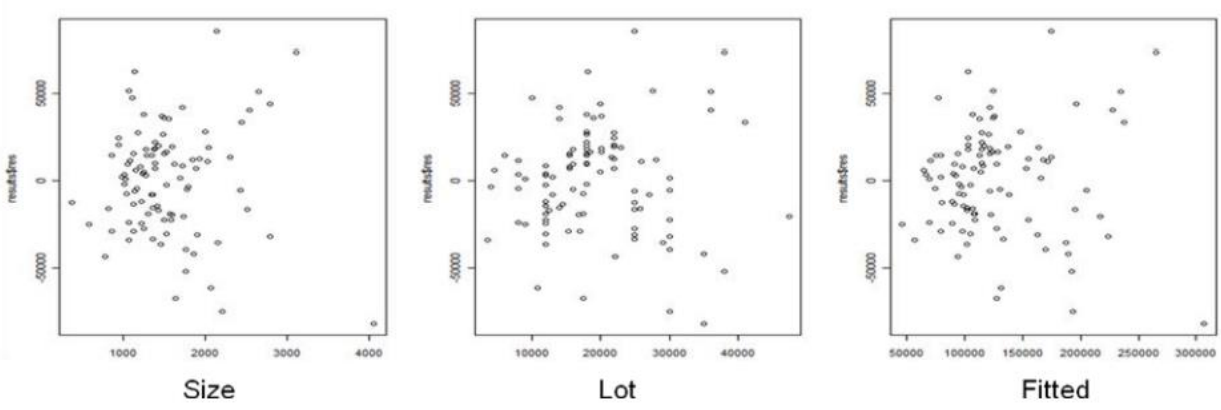
We can use residuals to study whether:

- The regression function is nonlinear.
- The error terms have non-constant variance.
- The error terms are not independent.
- There are outliers.
- The error terms are not normally distributed.

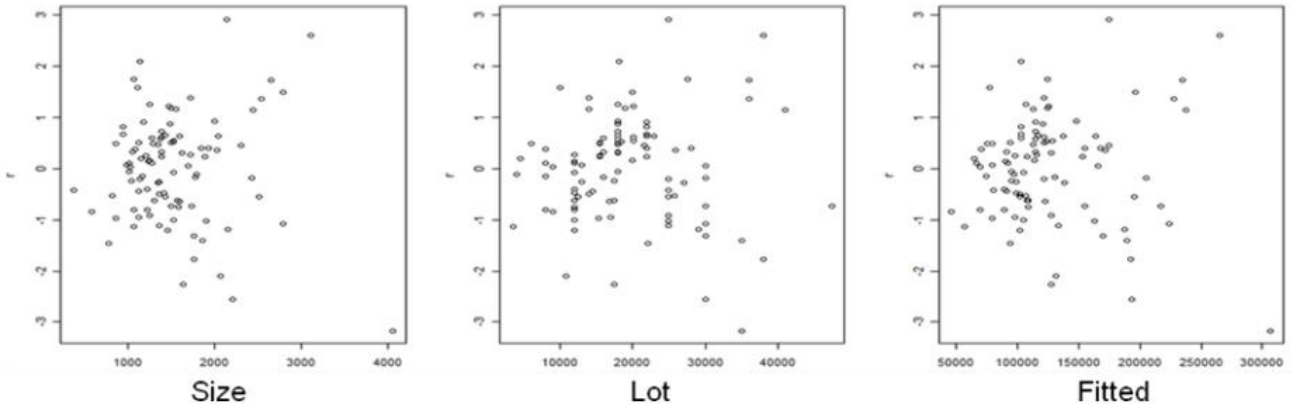
We can check for violations of these assumptions by making plots of the residuals, including:

- Plots of the residuals against the explanatory variable or fitted values.
- Histograms or boxplots of the residuals.
- Normal probability plots of the residuals.

We plot the residuals against both the predicted values and the explanatory variables. The residuals should be randomly scattered about 0 and the width should be equal throughout for the constant variance assumption to hold.

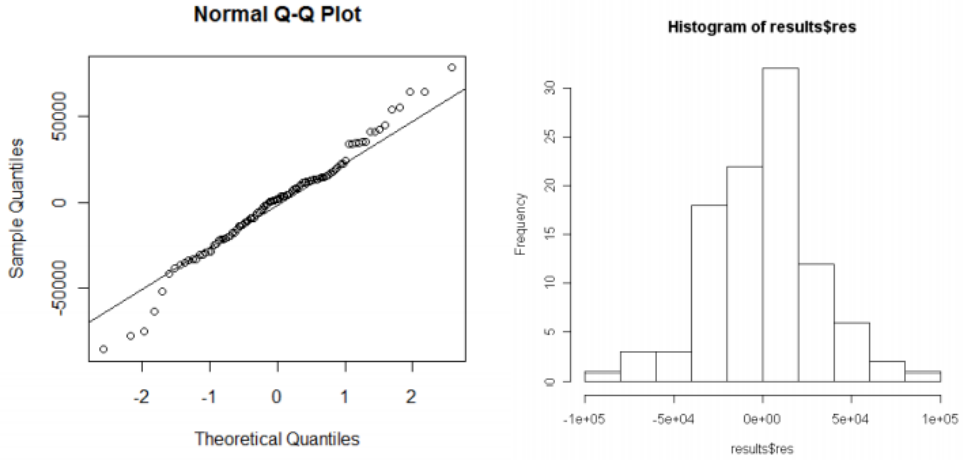


To make the same residual plots using the standard residuals type:

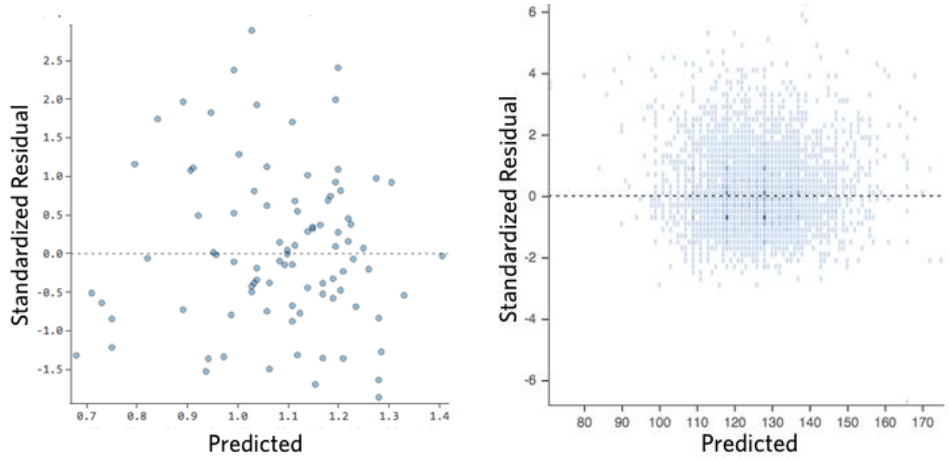


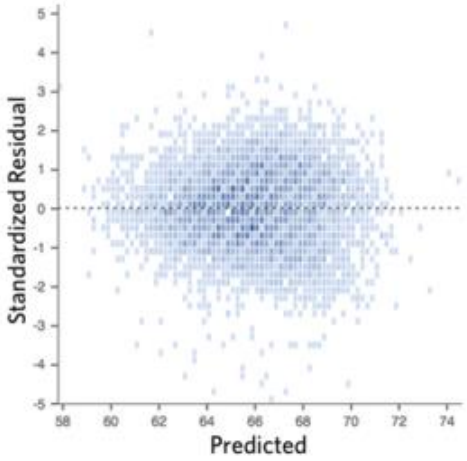
A Normal probability plot of the residuals can be used to check the normality assumption. Here each residual is plotted against its expected value under normality. To make normal probability plots, as well as a histogram, type:

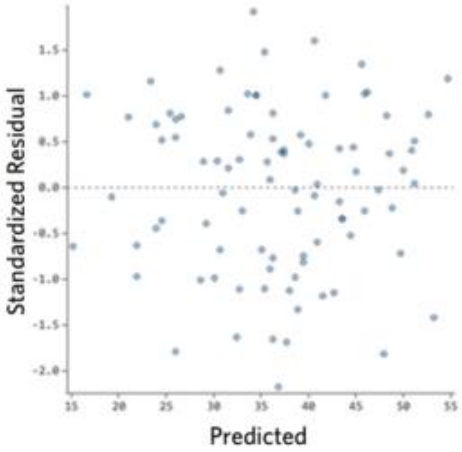
```
> qqnorm(results$res)
> qqline(results$res)
> hist(results$res)
```



Ideally your plots of the residuals look like one of these:



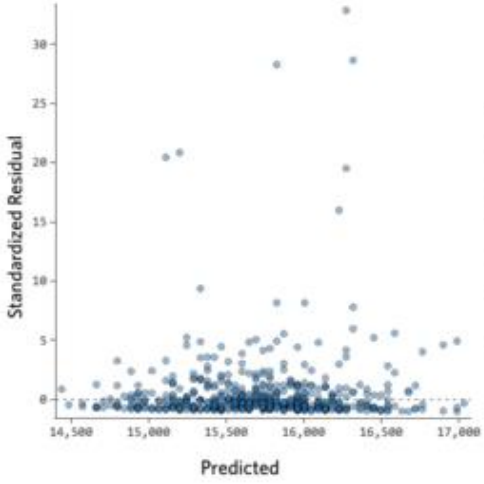


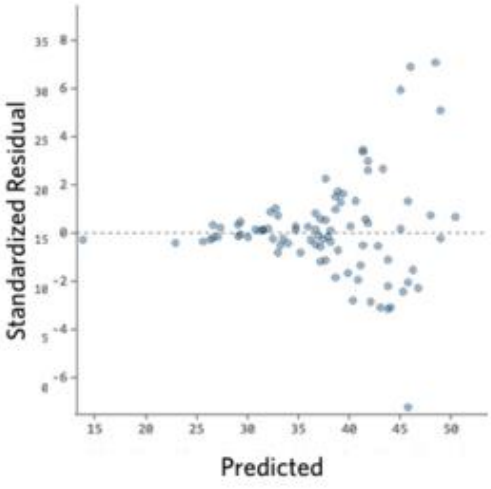


That is
1) they're pretty symmetrically distributed, tending to cluster towards the middle of the plot
2) they're clustered around the lower single digits of the y-axis (e.g., 0.5 or 1.5, not 30 or 150)
3) in general there aren't clear patterns

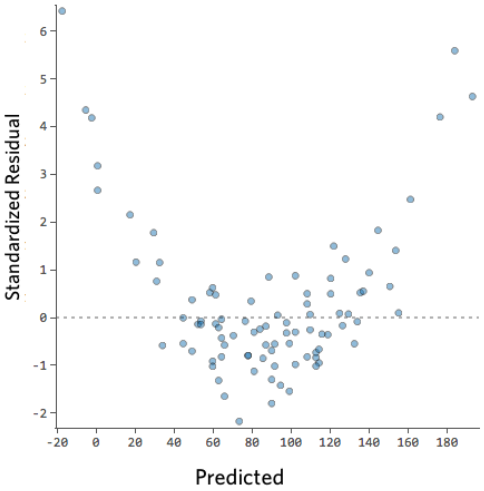
Here's some residual plots that don't meet those requirements:

Heteroscedasticity





Nonlinear



Outlier

